

Sec 7.1 and 7.2 – Integration Review

Basic Antiderivatives

Test yourself! Complete each of these antiderivative rules to see what you remember. Practice these rules so that they are automatic so you can recognize when to use them and focus your attention on the harder processes of integrals.

$$\text{For } n \neq -1, \int x^n dx = \underline{\hspace{2cm}} \quad \int \frac{1}{\cos^2(x)} dx = \underline{\hspace{2cm}}$$

$$\int \frac{1}{x} dx = \underline{\hspace{2cm}} \quad \int e^x dx = \underline{\hspace{2cm}}$$

$$\int \sin(x) dx = \underline{\hspace{2cm}} \quad \int \frac{1}{1+x^2} dx = \underline{\hspace{2cm}}$$

$$\int \cos(x) dx = \underline{\hspace{2cm}} \quad \int \frac{1}{\sqrt{1-x^2}} dx = \underline{\hspace{2cm}}$$

Using Algebra and Substitution to Evaluate Integrals

There are many methods of integration and we are only getting to a few in Calculus I! In all methods, you should think of algebra as a tool to let you make changes to the function you are integrating to make your work easier, as illustrated by these examples which only use the methods of antiderivatives and substitution.

$$\int (3x+3)(x+1) dx$$

$$\int \frac{4-3t^4+t^8}{t} dt$$

$$\int \frac{\sin(x)}{\cos(x)+3} dx$$

$$\int \frac{x+2}{1+x^2} dx$$

$$\int (x^3-1)^4 x^2 dx$$

$$\int \frac{4}{x^2} - \frac{3}{x^2+1} dx$$

$$\frac{d}{dx} \int_2^x t^3 e^t \, dt$$

$$\frac{d}{dw} \int_{-2}^{\sin(w)} e^{-x^2} \, dx$$

$$\int x \cos x \, dx$$

$$\int \cos^2 x \, dx$$

$$\int \ln(2+x) \, dx$$